

Streaming flow — SRT / NDI

Primary path (SRT)

```
VMix PCs (both nodes) --SRT--> Restreamer (mothership) --SRT (fan-out)--> BirdDog Play ×12 (one per theatre)
```

SRT is unicast-by-IP, so this path has no discovery dependency and should route cleanly over the tailnet.

Set the SRT payload size to 1128 bytes on every sender (VMix's SRT outputs and each Restreamer SRT output). SRT's default 1316B payload makes a 1360B IP packet, which is larger than Tailscale's 1280B tunnel MTU, so default-sized SRT packets would be fragmented (or dropped if DF is set) entering the tunnel. 1128B (6 × 188B MPEG-TS packets) gives a 1172B packet that fits. See `docs/bandwidth-analysis.md`.

Backup path (NDI)

```
BirdDog Central (mothership) --NDI--> BirdDog Play ×12 (one per theatre)
```

How Central gets the program — NDI redistribution (settled). Central works as an NDI *redistributor*: the source VMix node PC's native NDI output registers with the NDI Discovery Server, Central receives it over that node's own uplink (~130 Mbps against the node's Slate AX ceiling — estimated ~150-250 Mbps under Tailscale, unbenchmarked — a **fallback-only** load — pausing that node's VMix record ingest for the duration keeps it under the ceiling, the same lever the theatre side already uses), and Central re-sends it to the theatre PLAYS. Two researched facts pin this shape:

- **BirdDog Central is NDI-only on the input side** — its own user guide lists NDI sources, an NDI file-player, and an NDI re-transmitter; there is no SRT/RTMP ingest anywhere in it (Central 2.0 User Guide). So the feed into Central must already be NDI — it can't terminate the SRT program.
- **Restreamer can't be the NDI sender instead** — upstream FFmpeg removed NDI support in 2019 over a NewTek GPL violation and never reinstated it; datarhei's own FFmpeg build contains no NDI library and the maintainers have declined to ship it for licensing reasons (datarhei/restreamer discussion #391).

Remaining verification items (not design questions) are tracked as `docs/open-questions.md` #16. BirdDog Central was deliberately retained for this sender role when its fleet-management duties moved to Flock — see `docs/birddog-play-rationale.md`.

Bandwidth note: full NDI (not HX) at 1080p50 is ~125 Mbps per stream, roughly constant regardless of content — 12-22x SRT's figures, and without SRT's "static content is cheap" discount. Fine for a single theatre falling back — *provided that theatre's ATEM ISO ingest is paused for the duration*, since NDI plus

even the default camera + program ingest (~213 Mbps combined) is over the bottom of the theatre's own Slate AX Tailscale estimate (~150-250 Mbps) — and a real capacity question if this path ever has to carry all 12 theatres at once (e.g. a Restreamer failure). See `docs/bandwidth-analysis.md` for the full impact at per-router, per-VLAN, and mothership-NIC level.

Caveat, and the fix: NDI relies on mDNS multicast for discovery, which does not cross a Tailscale-routed boundary — Tailscale is point-to-point WireGuard and doesn't forward multicast/mDNS (still an open feature request, not something MagicDNS papers over). Plain local mDNS discovery will not find BirdDog Central across theatres on its own.

Plan: run an NDI Discovery Server on the mothership. NDI 5+ added this specifically for WAN/VPN scenarios — it's unicast TCP (default port **5959**), so it works fine across the tailnet with no multicast involved:

- Run the discovery server as a small service on the mothership, alongside BirdDog Central.
- Point every NDI participant at that server's IP: the VMix node PCs (senders of the program feed Central redistributes), BirdDog Central (receiver of that feed *and* sender toward the theatres), and all 12 BirdDog Play units (receivers). On Windows/macOS this is the NDI Tools Access Manager "Advanced" tab; on Linux it's `~/.ndi/ndi-config.v1.json`.
- Once a sender has a discovery server configured, it stops using mDNS entirely — it's visible only to finders pointed at the same server.
- Tailscale MagicDNS can supply a stable hostname for the discovery server's address instead of hardcoding its Tailscale IP, if the receiving device's config field accepts a hostname — a convenience, not a substitute for the discovery server itself.

Confirmed on BirdDog Play: BirdUI (the PLAY's web admin — browse to its IP, default password `birddog`) has a Network panel with NDI-specific settings including a Discovery Server toggle. Switch it ON, enter a comma-delimited list of Discovery Server IP address(es), click APPLY. BirdDog's own docs describe this as the mechanism for locating NDI sources "on different subnets" — exactly this cross-theatre case — and it supports multiple servers for failover. Sources: BirdDog PLAY User Guide, BirdUI User Guide. Change the default admin password before the event — BirdUI grants full device config access.

See `diagrams/streaming-flow.svg` — primary path solid, backup path dashed. For why BirdDog Play is the theatre-side receiver at all (vs. a laptop

- VLC), see `docs/birddog-play-rationale.md`.